Fiscal Year:	FY 2022	**Task Last Updated:**	FY 03/25/2022
PI Name:	Rosenberg, Marissa Ph.D.		
Project Title:	Exploration Exercise Treadmill Requirements		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2021	**End Date:**	06/30/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	End date changed to 06/30/2022 as Dr. Karina Marshall-Goebel took over this project in June 2022. Original end date was 09/30/2027.		
Key Personnel Changes/Previous PI:	NOTE: P.I. changed to K. Marshall-Goebel effective 06/27/2022. For subsequent reporting, please see "Exploration Exercise Treadmill Requirements (PI: Marshall-Goebel)". (Ed. 8/4/2022)		
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Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Exploration class missions including Artemis, Gateway, and beyond, are required to have an exercise device that is lightweight and has a small footprint. This has resulted in the development of exercise devices that are motorized and flywheel-based to provide both aerobic and resistive training on one platform. These devices provide a variety of full body resistance exercise options, and aerobic rowing and cycling, but no treadmill. Because a treadmill has been available for use on the International Space Station (ISS) since 2001, we do not understand what the effect of removing the treadmill will have on muscle performance, aerobic fitness, bone health, sensorimotor performance, or psychological health. A retrospective study was performed to quantify the association between total exercise (including specific contributions of treadmill exercise) and functional performance upon landing. Due to the relatively small variation in treadmill usage inflight, there is a clear need to do a controlled study, where crewmembers will not use the treadmill, to determine the effect of no treadmill use during spaceflight on performance decrements immediately after landing, as well as the recovery time back to baseline for bone health, muscle performance, aerobic fitness, sensorimotor performance, and psychological health. Specific Aims: 1) We aim to quantify the effect of cycle ergometry and resistive exercise only, throughout an entire spaceflight mission on muscle performance, aerobic fitness, bone health, sensorimotor performance, or psychological health by removing the treadmill portion of exercise on ISS, and replacing it with additional time on the ISS cycle ergometer (CEVIS), as prescribed by the astronaut trainers, while keeping exercise on the Advanced Resistive Exercise Device (ARED) the same. Gateway, the human landing system (HLS), and Orion plans do not include a treadmill; however, it is still possible to have treadmill capability for future exploration missions such as Martian transits. Therefore, it is critical to determine the importance of the treadmill to maintaining human performance capabilities as soon as possible to define evidence-based exercise system requirements for exploration vehicles. Inaccurate or inadequate countermeasure requirements can result in downstream impacts on cost, schedule, operations, and potentially crew health and performance. 2) We aim to quantify the effectiveness of the Exploration Exercise Device (EED) during an entire spaceflight mission on bone, muscle, aerobic, and sensorimotor health and performance by having crewmembers utilize the EED exclusively. The one exception to this would be the quantification of peak oxygen uptake (V02peak), which will be completed on CEVIS periodically and an isometric mid-thigh pull (IMTP) performed on ARED. Future exploration missions plan to use a similar EED; thus, we must validate its effectiveness as a countermeasure providing sufficient protection for bone, muscle, aerobic, and sensorimotor health and performance. This validation is critical to ensure that we are adequately protecting crew on future long-duration missions.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. Exploration class missions including Artemis, Gateway, and beyond, are required to have an exercise device that is lightweight and has a small footprint. This has resulted in the development of exercises devices that are motorized and flywheel-based to provide both aerobic and resistive training on one platform. These devices provide a variety of full body resistance exercise options as well as aerobic rowing and cycling, but no treadmill. Without a treadmill, there is no means of ambulating, or reinforcing the motor pattern of walking. This study aims to quantify the effect of no treadmill usage during an entire spaceflight mission on bone, muscle, aerobic, and sensorimotor health and performance.
Research Impact/Earth Benefits:	
Task Progress:	NOTE: P.I. changed to K. Marshall-Goebel effective 06/27/2022. For subsequent reporting, please see "Exploration Exercise Treadmill Requirements (PI: Marshall-Goebel)". (Ed. 8/4/2022). New project for FY2022.
Bibliography Type:	Description: (Last Updated: 03/22/2017)